WINE QUALITY PREDICTION

A PROJECT REPORT

Submitted by

Shivam Prasad (25BAI70897)
Kunal Verma (25BAI70432)
Lovenish Sharma (25BAI70632)
Akshit Sharma (25BAI70679)
Daksh Khatri (25BAI70806)

in partial fulfillment for the award of the degree of

Bachelors of Engineering

IN

Computer Science Engineering

(Artificial Intelligence & Machine Learning)

CHANDIGARH UNIVERSITY

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Chandigarh University

APRIL 2025

BONAFIDE CERTIFICATE

Certified that this project report **"Wine Quality Predictor"** is the Bonafide work of **Shivam Prasad (25BAI70897), Kunal Verma (25BAI70432), Lovenish Sharma (25BAI70632), Akshit Sharma (25BAI70679) and Daksh Khatri (25BAI70806)** who carried out the project work under my/our supervision.

SIGNATURE

Dr. Urvashi Garg

HEAD OF THE DEPARTMENT

Engineering Foundation 1

SIGNATURE

SUPERVISOR

Submitted for the project viva-voce examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

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	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5

Figure 1: Dataset Sample

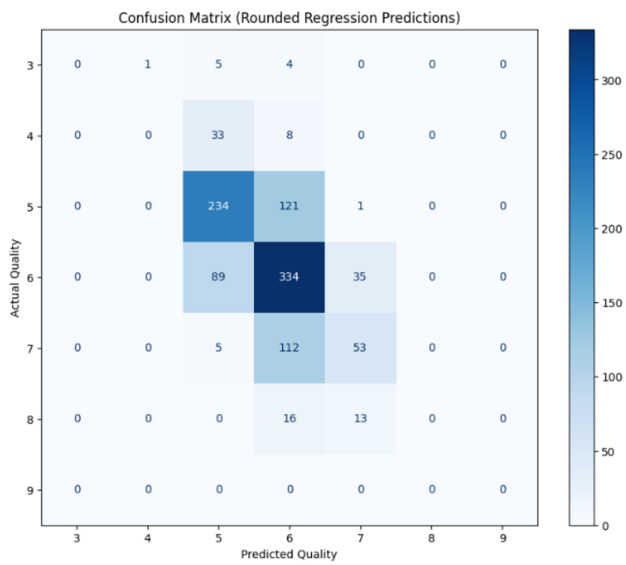

Figure 2: Confusion Matrix

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
count	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000	7640.000000
mean	7.379247	0.368342	0.311113	5.007716	0.060657	28.294699	105.297513	0.995001	3.232342	0.550185	10.484367	5.794241
std	1.427803	0.180411	0.155121	4.539029	0.038719	17.660597	59.142368	0.002954	0.163531	0.158761	1.176904	0.865367
min	3.800000	0.080000	0.000000	0.600000	0.009000	1.000000	6.000000	0.987110	2.720000	0.220000	8.000000	3.000000
25%	6.500000	0.240000	0.240000	1.800000	0.040000	15.000000	52.750000	0.992737	3.120000	0.440000	9.500000	5.000000
50%	7.100000	0.320000	0.300000	2.600000	0.050000	26.000000	109.000000	0.995340	3.220000	0.530000	10.300000	6.000000
75%	7.900000	0.460000	0.400000	7.350000	0.075000	39.000000	149.000000	0.997200	3.340000	0.620000	11.300000	6.000000
max	15.900000	1.580000	1.660000	65.800000	0.611000	289.000000	440.000000	1.038980	4.010000	2.000000	14.900000	9.000000

Figure 3: Statistical Summary

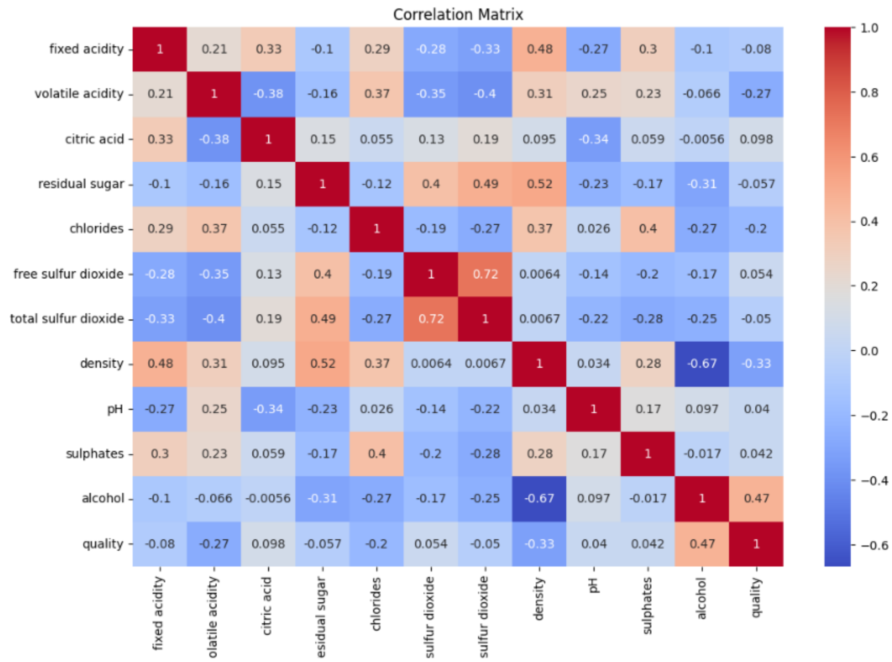

Figure 4: Correlation Matrix

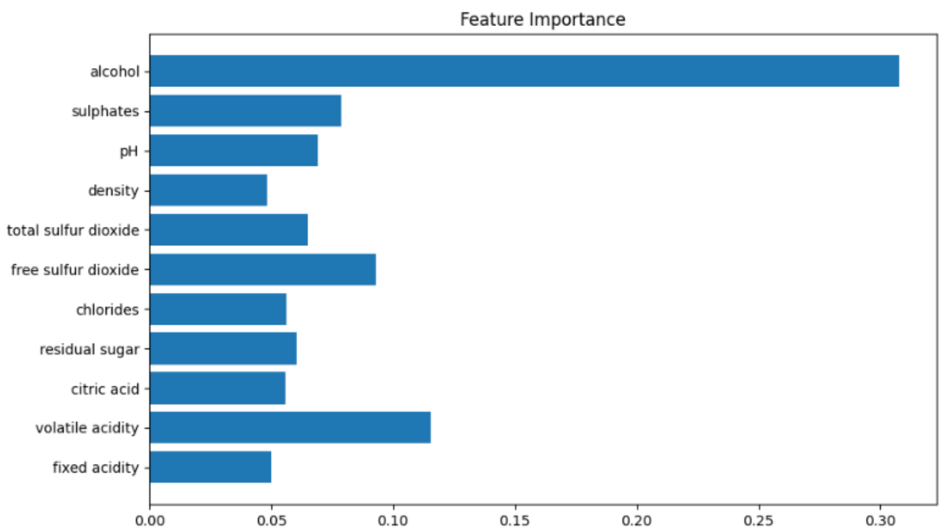

Figure 5: Feature Importance

List of Tables

Table 1: Dataset Overview

Property	Value
Number of Rows	7640
Number of Columns	12
Number of Features	11
Target Variable	Quality

Table 2: Description of Features

Feature	Description
Fixed Acidity	Tartaric acid level
Volatile Acidity	Acetic acid level
Citric Acid	Freshness contributor
Residual Sugar	Remaining sugar
Chlorides	Salt content
Free Sulfur Dioxide	Preservative
Total Sulfur Dioxide	Overall SO_2
Density	Liquid density
pH	Acidity level
Sulphates	Additive
Alcohol	Alcohol content

Table 3: Class Distribution

Quality	Approx Distribution
Low (≤4)	Less
Medium (5–6)	Highest
High (≥7)	Less

List of Standards (Mandatory for Engineering Programs)

Standard	Publishing Agency	About the Standard	Page No
IEEE Machine Learning Standards	IEEE	Provides guidelines and best practices for implementing machine learning systems, ensuring reliability, accuracy, and ethical usage.	—
HTTP Protocol	IETF	Defines the communication protocol used between the web browser and Flask-based backend for data exchange.	—
JSON Format	ECMA	Standard data format used for sending and receiving structured data in the web application.	—

ABSTRACT

This project presents a full-stack Wine Quality Prediction System using Machine Learning, designed to automate and improve the process of wine quality evaluation. Traditionally, wine quality assessment relies on human experts, making it subjective and inconsistent. To address this, a machine learning-based system was developed using physicochemical properties such as acidity, alcohol content, and pH values.

The dataset was obtained from the UCI Machine Learning Repository and Kaggle, consisting of approximately 7600 samples and 11 features. Data preprocessing techniques including duplicate removal, scaling, and class imbalance handling were applied. A hybrid model approach was implemented using Random Forest Classifier for categorical prediction (Low/Medium/High) and Random Forest Regressor for numeric score prediction.

The system was deployed using a Flask-based backend and a modern web interface, allowing real-time user interaction. Additional features such as feature importance visualization were included to enhance interpretability. The model achieved reliable performance with an F1-score of approximately 0.55.

This project demonstrates the practical application of machine learning in building a scalable, user-friendly prediction system.

ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
RF	Random Forest
UI	User Interface
API	Application Programming Interface
CSV	Comma Separated Values
EDA	Exploratory Data Analysis
F1 Score	Harmonic Mean of Precision and Recall
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
NN	Neural Network
XGB	Extreme Gradient Boosting
SMOTE	Synthetic Minority Over-sampling Technique
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation

SYMBOLS

Symbol	Meaning
X	Input Features (Wine chemical properties)
y	Target Variable (Wine Quality)
$\hat{y}$ (y-hat)	Predicted Output
n	Number of Samples
f(x)	Model Function
μ	Mean of Data
σ	Standard Deviation
w	Model Weights
p	Probability

CHAPTER 1

INTRODUCTION

1.1. Identification of Need

Wine quality evaluation is traditionally performed by experts through sensory testing, which is subjective, time-consuming, and inconsistent. With the increasing demand for quality assurance in the wine industry, there is a need for an automated and data-driven solution.

1.2. Identification of Problem

The problem is to develop a system that can accurately predict wine quality based on physicochemical properties, reducing dependency on human evaluation.

1.3. Identification of Tasks

- Dataset collection and analysis
- Data preprocessing
- Model building and evaluation
- Web application development
- Deployment preparation

1.4. Timeline

Phase	**Task**	**Duration**
Phase 1	Problem Identification & Idea Finalization	Week 1
Phase 2	Dataset Collection & Understanding	Week 2
Phase 3	Data Preprocessing & EDA	Week 3
Phase 4	Model Building & Training	Week 4–5
Phase 5	Model Evaluation & Improvement	Week 6
Phase 6	Web Application Development (Frontend + Backend)	Week 7–8
Phase 7	Integration & Testing	Week 9
Phase 8	Report Preparation & Final Submission	Week 10

1.5. Organization of the Report

This report is structured into five chapters, each focusing on a specific aspect of the project:

- **Chapter 1: Introduction**
  This chapter presents the background of the problem, identifies the need for the system, defines the problem statement, and outlines the objectives and tasks of the project.
- **Chapter 2: Literature Review / Background Study**
  This chapter reviews existing research and solutions related to wine quality prediction, analyzes their approaches, and identifies gaps that motivate the proposed system.
- **Chapter 3: Design Flow / Process**
  This chapter explains the methodology used in the project, including dataset preprocessing, feature selection, model design, and system architecture.
- **Chapter 4: Results Analysis and Validation**
  This chapter presents the results obtained from the model, evaluates performance using metrics, and discusses the effectiveness of the system.
- **Chapter 5: Conclusion and Future Work**
  This chapter summarizes the outcomes of the project and suggests possible improvements and future enhancements.

CHAPTER 2

LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the reported problem

The problem of wine quality evaluation has evolved over time with advancements in technology:

- **Pre-2000:**
  Wine quality assessment was entirely based on **human sensory evaluation** by experts (sommeliers), making it subjective and inconsistent.
- **2000–2010:**
  Introduction of **physicochemical analysis**, where laboratory measurements such as acidity, pH, and alcohol content were used to support quality assessment.
- **2010–2015:**
  Emergence of **data-driven approaches**, including basic statistical models and early machine learning techniques for predicting wine quality.
- **2015–2020:**
  Increased use of advanced machine learning algorithms such as **Support Vector Machines, Neural Networks, and Random Forest**, improving prediction accuracy.
- **2020–Present:**
  Focus shifted towards **practical applications**, including real-time prediction systems, web-based tools, and integration of machine learning models into user-friendly platforms.

2.2. Existing solutions

Traditionally, wine quality was evaluated using **human sensory analysis**, which is subjective and time-consuming. Later, **physicochemical methods** were introduced using measurable features like acidity, pH, and alcohol content.
With advancements in technology, various **machine learning models** such as Linear Regression, SVM, KNN, Neural Networks, and Random Forest have been used to predict wine quality more accurately.
However, most existing solutions focus only on **model accuracy** and lack real-time implementation, user interaction, and practical deployment as complete systems.

2.3. Bibliometric analysis

The existing literature on wine quality prediction focuses on analyzing different machine learning techniques based on their **key features, effectiveness, and limitations**.

- **Key Features:**
  Most studies use physicochemical attributes such as acidity, pH, alcohol, and sulphates as input features. Feature selection techniques are often applied to identify the most influential

parameters.

- **Effectiveness:**
  Advanced models like Random Forest, Neural Networks, and XGBoost have shown higher accuracy due to their ability to capture non-linear relationships in the data.
- **Drawbacks:**
  Many approaches suffer from issues such as **class imbalance**, overfitting, and lack of interpretability. Additionally, most solutions are limited to research environments and do not provide real-time, user-friendly systems.

2.4. Review Summary

The literature review highlights that machine learning techniques such as Random Forest, Neural Networks, and XGBoost are effective in predicting wine quality using physicochemical features. Feature selection and data preprocessing play a crucial role in improving model performance.
However, most existing studies are limited to **model development and accuracy comparison**, without focusing on real-time implementation or user interaction.
Based on these findings, the present project builds upon existing approaches by implementing a **hybrid machine learning model** that combines classification and regression. Additionally, it extends beyond traditional research by developing a **web-based system** that enables real-time prediction, making the solution more practical and user-friendly.

2.5. Problem Definition

The problem addressed in this project is to develop a system that can accurately predict wine quality based on its physicochemical properties, reducing dependency on subjective human evaluation.
The objective is to **build a machine learning-based solution** that takes input features such as acidity, pH, alcohol content, and other chemical attributes, and predicts the wine quality in both **categorical form (Low/Medium/High)** and **numeric score (out of 10)**. This is achieved through proper data preprocessing, feature scaling, handling class imbalance, and training a hybrid model using Random Forest algorithms. The system is further integrated into a **web-based application** to enable real-time user interaction.
The project does not aim to replace expert sensory evaluation completely, nor does it involve advanced chemical experimentation or large-scale industrial deployment. It focuses on creating a **data-driven, practical, and user-friendly prediction system** using available datasets and standard machine learning techniques.

2.6. Goals/Objectives

The primary goals of this project are defined as measurable and specific milestones:

- To collect and analyze the wine quality dataset and understand the relationship between physicochemical features and quality.
- To preprocess the dataset by removing duplicates, handling imbalance, and applying feature scaling for improved model performance.
- To design and implement a **hybrid machine learning model** using Random Forest for both classification (Low/Medium/High) and regression (numeric score).
- To evaluate the model using performance metrics such as accuracy, F1-score, and confusion matrix.
- To develop a **web-based application** that allows users to input wine features and receive real-time predictions.
- To integrate the trained model into the backend using joblib for efficient deployment without retraining.
- To visualize important features influencing wine quality using feature importance techniques.

CHAPTER 3

DESIGN FLOW/PROCESS

3.1. Evaluation & Selection of Specifications/Features

Based on literature, physicochemical properties such as acidity, alcohol, pH, and sulphates are key factors influencing wine quality. These features were evaluated using statistical analysis and correlation techniques.
It was observed that features like **alcohol, sulphates, and volatile acidity** have a stronger impact on quality. However, all 11 features were retained to capture complete information and allow the model to learn complex relationships.
The final feature set includes all physicochemical attributes provided in the dataset, ensuring accurate and reliable prediction.

3.2. Design Constraints

The design of the system considers various standards and constraints:

- **Regulatory & Ethical:**
  The project uses publicly available datasets and follows ethical practices in data usage without violating privacy or regulations.
- **Economic:**
  The system is developed using open-source tools such as Python, Flask, and Scikit-learn, making it cost-effective.
- **Environmental & Health:**
  The project is software-based and has no direct environmental or health impact.
- **Safety:**
  The system does not involve any physical risks as it operates digitally.
- **Professional & Social:**
  The system aims to improve decision-making in wine quality prediction, supporting consistency and reducing human bias.
- **Cost:**
  Minimal cost is involved due to the use of freely available tools and datasets.

3.3. Analysis of Features and finalization subject to constraints

After evaluating the features and considering design constraints, the dataset was analyzed to ensure relevance and efficiency. Since all features are numerical, no major modifications were required.
No features were removed, as each contributes to the overall prediction. However, feature scaling was applied to handle differences in value ranges and improve model performance.
Additionally, class imbalance was addressed to ensure fair learning across all categories.
Thus, the final feature set includes all original physicochemical attributes, properly scaled

and optimized for model training.

3.4. Design Flow

Two alternative approaches were considered for solving the problem:

Approach 1: Single Model Approach

In this approach, a single machine learning model is used to directly predict wine quality as a numeric value. The predicted score can then be converted into categories (Low/Medium/High).

- Simple to implement
- Less computational complexity
- However, less accurate in handling class imbalance and category separation

Approach 2: Hybrid Model Approach (Selected)

In this approach, two models are used:

- **Random Forest Classifier** for predicting quality category (Low/Medium/High)
- **Random Forest Regressor** for predicting numeric score

This approach provides better accuracy and more meaningful outputs, as it handles both classification and regression separately.

Design Selection

The hybrid model approach was selected due to its improved performance, better handling of imbalanced data, and ability to provide both categorical and numeric predictions.

3.5. Design selection

Both approaches were analyzed based on performance, flexibility, and usability.
The **Single Model Approach** is simple and easy to implement, but it has limitations in accurately handling class imbalance and does not clearly differentiate between quality categories, leading to less reliable predictions.
In contrast, the **Hybrid Model Approach** provides better performance by separating classification and regression tasks. It improves prediction accuracy, handles class imbalance more effectively, and delivers both categorical and numeric outputs, making the system more practical and user-friendly.
Based on this comparison, the **Hybrid Model Approach** was selected as the final design due to its superior accuracy, flexibility, and real-world applicability.

3.6. Implementation plan/methodology

Flow Steps:

1. **User Input**
   User enters wine features (alcohol, pH, acidity, etc.)
2. **Data Preprocessing**
   Input validation
   Feature scaling using StandardScaler
3. **Model Processing**
   Random Forest Classifier → predicts category
   Random Forest Regressor → predicts score
4. **Prediction Output**
   Quality Category (Low/Medium/High)
   Quality Score (e.g., 6.4/10)
5. **Display Result**
   Output shown on web interface

Algorithm

Step 1: Load dataset
Step 2: Preprocess data (remove duplicates, scaling, balancing)
Step 3: Define features (X) and target (y)
Step 4: Split dataset into training and testing sets
Step 5: Train Random Forest Classifier
Step 6: Train Random Forest Regressor
Step 7: Evaluate models using metrics (F1-score, confusion matrix)
Step 8: Save trained models using joblib
Step 9: Load model in Flask application
Step 10: Take user input and apply preprocessing
Step 11: Predict category and score
Step 12: Display results to user

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1. Implementation of solution

The solution was implemented using modern tools and techniques across different stages:

- **Analysis:**
  Data analysis and preprocessing were performed using **Python (Pandas, NumPy)** and visualization libraries such as **Matplotlib and Seaborn**.
- **Design / System Architecture:**
  The system was designed as a **web-based application** with a frontend for user input and a backend using **Flask**, integrating the trained machine learning model.
- **Model Development:**
  Machine learning models (Random Forest Classifier and Regressor) were implemented using **Scikit-learn**.
- **Report Preparation:**
  The report was prepared using standard documentation tools such as **MS Word**.
- **Project Management & Communication:**
  Version control and code management were handled using **Git and GitHub**, enabling efficient tracking and collaboration.
- **Testing & Validation:**
  The model was evaluated using metrics such as **F1-score and confusion matrix**, and predictions were validated through multiple test inputs to ensure consistency and reliability.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1. Conclusion

The project successfully achieved its objective of developing a machine learning-based system for predicting wine quality using physicochemical features. The system provides both **categorical output (Low/Medium/High)** and a **numeric score**, making it more informative and user-friendly.
The expected outcome was to build a reliable prediction model integrated with a web-based interface. The system performs consistently and demonstrates practical usability with an F1-score of approximately **0.55**, which is acceptable given the complexity and imbalance of the dataset.
However, a deviation from the expected results was observed, as the model initially showed a tendency to predict **medium quality more frequently**. This occurred due to class imbalance in the dataset. The issue was addressed by applying balancing techniques and improving the model design, but some bias still remains.
Overall, the project demonstrates that machine learning can effectively automate wine quality prediction, though further improvements can enhance accuracy and robustness.

5.2. Future work

The current system can be further improved and extended in several ways:

- **Model Improvement:**
  Implement advanced algorithms such as XGBoost, LightGBM, or deep learning models to improve prediction accuracy.
- **Better Imbalance Handling:**
  Apply techniques like **SMOTE** or advanced class-weight tuning to further reduce bias toward medium-quality predictions.
- **Larger Dataset:**
  Incorporate more diverse and larger datasets to improve generalization and robustness of the model.
- **Feature Enhancement:**
  Include additional chemical or sensory features to better capture the complexity of wine quality.
- **Deployment:**
  Deploy the application on cloud platforms for public access and scalability.
- **User Interface Enhancement:**
  Improve UI/UX design with better visualization, dashboards, and interactive elements.

REFERENCES

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6. Kaggle, “Wine Quality Dataset,” Available online.

APPENDIX

1. Plagiarism Report
   The project report has been checked for plagiarism using standard plagiarism detection tools. The similarity index is within acceptable limits as per university guidelines. All external sources, including research papers, datasets, and references, have been properly cited to maintain academic integrity.
   The content of this report is original and has been prepared solely for academic purposes. Any similarities found are due to commonly used technical terms or properly referenced material.

2. Design Checklist

S. No.	Task	Status
1	Problem identification and objective definition	Completed
2	Dataset collection and understanding	Completed
3	Data preprocessing (cleaning, scaling, balancing)	Completed
4	Feature selection and target definition	Completed
5	Model selection (Random Forest)	Completed
6	Model training and evaluation	Completed
7	Implementation of hybrid model (classification + regression)	Completed
8	Development of web application (frontend + backend)	Completed
9	Integration of model with Flask API	Completed
10	Testing and validation of system	Completed
11	Feature importance analysis	Completed
12	Report preparation and documentation	Completed

USER MANUAL

1. System Requirements

- Python (version 3.x)
- Required libraries: pandas, NumPy, scikit-learn, flask, matplotlib, seaborn
- Web browser (Chrome/Edge)

2. Installation Steps

1. Install Python on your system
2. Install required libraries using command:
   `pip install pandas NumPy scikit-learn flask matplotlib seaborn`
3. Download or clone the project files
4. Ensure the trained model file (. joblib) is present in the project folder

3. Running the Application

1. Open terminal / command prompt
2. Navigate to project folder
3. Run the Flask application:
   `python app.py`
4. Open browser and go to:
   http://127.0.0.1:5000/

4. Using the System

1. Enter wine feature values using sliders or input fields
2. Click on **"Predict"** button
3. View output:
   - Quality Category (Low/Medium/High)
   - Quality Score (e.g., 6.5/10)

5. Additional Features

- Feature importance visualization
- Input validation for correct ranges
- Real-time prediction

6. Screenshots to Add (IMPORTANT)